Project: Worked examples
Title: Pipe pressure drop and pump duty
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Friction and fitting losses through series pipe segments at a given flow, the static head, the system curve, the duty point against a pump curve entered as points, the affinity-law shift for a speed or impeller change, and the NPSH available at the pump suction.

WHAT IT CALCULATES	RELATION
Design flow from the heat carried, when a heat load is given in place of a flow	$Q = \Phi/(\rho\, c_p\, \Delta T)$
Mean velocity in the pipe	$v = Q/(\pi D^2/4)$
Reynolds number, which decides laminar or turbulent flow	$Re = \rho v D/\mu$
Friction factor for laminar flow (Re < 2300)	$f = 64/Re$
Friction factor for turbulent flow, Colebrook–White by fixed-point iteration (at most 100 steps) from the Swamee–Jain value, which also cross-checks it	$1/\sqrt{f} = -2\log_{10}(\varepsilon/3.7D + 2.51/(Re\sqrt{f}))$
Head lost in the pipe and its fittings	$h_f = f\,(L + L_{eq})/D \cdot v^2/2g + K\, v^2/2g$
System curve, with the friction of a turbulent or transitional segment and every fitting loss scaled from the design flow as Q^2, and the friction of a laminar segment (Re < 2300, $h_f \propto Q$) as Q (a closed circuit has $\Sigma dz = 0$, the rise in the flow being recovered in the return)	$H_{sys}(Q) = \Sigma dz + h_{out} + \Sigma h_{f,lam}\cdot(Q/Q_o) + \Sigma h_{f,other}\cdot(Q/Q_o)^2$
Water held in the pipes	$V_{pipe} = \Sigma\, \pi/4 \cdot D^2 \cdot L$
Duty point, where the pump and system curves cross (every crossing is found by a scan and refined by bisection; the stable one, where the pump curve falls through the system curve, nearest the design flow is reported)	$H_{pump}(Q) = H_{sys}(Q)$
Pump affinity laws for a change of speed n	$Q \propto n$; $H \propto n^2$. For a change of impeller diameter D, by the choice: a trimmed impeller of the same casing, $Q \propto D$ and $H \propto D^2$ (the default, the usual practice for a trim); or a geometrically similar pump scaled in all dimensions, $Q \propto D^3$ and $H \propto D^2$. Speed and diameter combine as $Q \propto n\cdot D\text{^}a$, $H \propto n^2\cdot D^2$, a = 1 or 3

WHAT IT CALCULATES	RELATION
Net positive suction head available, to compare with what the pump requires	$NPSH_a = p_{atm}/\rho g + z_s - h_{f,suction} - p_v/\rho g$, with p_v from the water correlation or, for a custom fluid, from the vapour pressure entered. The friction $h_{f,suction}$ is that of the segments marked 'suction' only, at the larger of the design flow and the duty flow the pump runs at
Shaft torque for the motor	$T = P/\omega$ with P the larger of the design-flow shaft power and the duty-point shaft power, at the speed n_{pump}; when a speed or diameter ratio shifts the duty, the shifted duty power at the shifted speed $n_{pump} \cdot n_2/n_1$

Basis: Flow of fluids through valves, fittings and pipe[1]; Internal flow: friction factor, minor losses, pump duty[2](Ch. 8); Flow of fluids in pipes and ducts (principles)[3](Section 4); Density of liquid water[4].

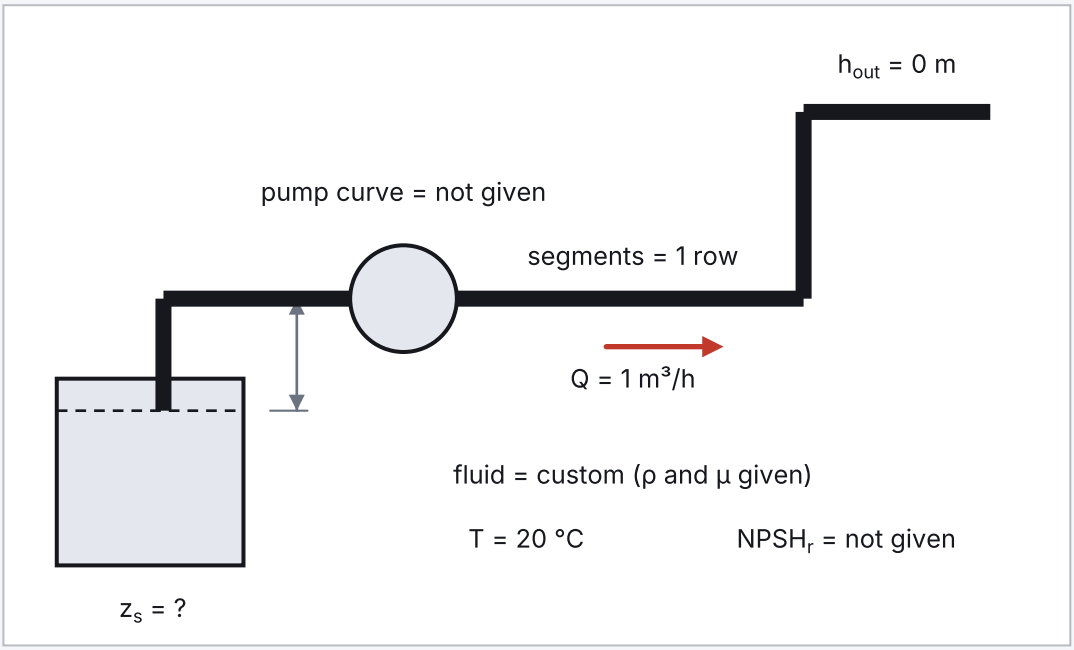


Figure 1 — A pump lifting from a suction level through the pipe segments as listed (three drawn) to an outlet that needs a pressure head. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
fluid	Fluid	custom (ρ and μ given)		spec: Oil datasheet ISO VG 680 at 20 °C	any
T	Fluid temperature (mean; link B6 t_m for a heating loop)	20	°C	assumption: Ambient 20 °C.	0 … 100 °C
ρ_{custom}	Density (custom fluid)	888	kg/m³	spec: Oil datasheet ISO VG 680	500 … 2000 kg/m³
μ_{custom}	Dynamic viscosity (custom fluid)	800	mPa·s	spec: Oil datasheet ISO VG 680 at 20 °C	0.1 … 1000 mPa·s
Q	Design flow rate (blank when a heat load gives it)	1	m³/h	spec: Transfer pump duty	0.01 … 100000 m³/h

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
segments	Pipe segments in series	1 row(s)		drawing: DRG-L-020 rev A	any
circuit	Circuit (a closed flow-and-return loop has no static lift: the rises entered are ignored)	open (the pump lifts the water by the rises entered)		default	any
h_{out}	Pressure head required at the outlet	0	m	assumption: Open discharge.	0 … 1000 m
v_{max}	Velocity limit (warning)	3	m/s	assumption: Practice limit.	0.5 … 10 m/s
n_{ratio}	Speed ratio n_2/n_1 (affinity)	1		default	0.2 … 2
D_{ratio}	Impeller diameter ratio D_2/D_1 (affinity; the law is chosen below)	1		default	0.7 … 1.1
D_{law}	Law for the diameter ratio: a trimmed impeller of the same casing ($Q \propto D$, $H \propto D^2$) or a geometrically similar pump ($Q \propto D^3$, $H \propto D^2$)	impeller trim ($Q \propto n \cdot D$, $H \propto n^2 \cdot D^2$)		default	any
η_{pump}	Pump efficiency at duty (for shaft power)	0.7		default	0.2 … 0.95
p_{atm}	Atmospheric pressure (NPSH)	101.325	kPa	default	50 … 110 kPa
z_s	Static head at the pump suction (+ flooded, − lift)	0	m	default	-10 … 100 m

TABLE ROW	VALUES
segments[1]	name = transfer line; L = 40 m; D = 50 mm; eps = 0 mm; K = 0; L_eq = 0 m; dz = 0 m; side = discharge

3. Assumptions

A1. Steady, incompressible, fully developed flow; all segments carry the full design flow in series. (Darcy–Weisbach.)
A2. Fitting losses scale with velocity head (K method) or add pipe length (equivalent-length method); both may be used on one segment. (Crane TP-410 practice.)
A3. The system curve scales the friction of turbulent and transitional segments and all fitting losses with Q^2 from the design flow (the friction factor is held at its design-flow value); the friction of a laminar segment scales with Q; the static part is independent of flow. ($H_{sys} = H_{static} + k_1 Q + k_2 Q^2$.)
A4. Water properties are from published correlations at atmospheric pressure (0–100 °C); any other fluid is entered with its own density and viscosity and their provenance. (Kell 1975; Vogel fit; Hyland & Wexler 1983.)
A5. T (Fluid temperature (mean; link B6 t_m for a heating loop)) is taken as 20 degC. (Ambient 20 °C.)
A6. h_{out} (Pressure head required at the outlet) is taken as 0 m. (Open discharge.)
A7. v_{max} (Velocity limit (warning)) is taken as 3 m/s. (Practice limit.)

4. Working

[5] $\rho = \rho_{custom}$
$= 888\ kg/m^3$
$= 888\ kg/m^3$
Density (as entered)

[6] $\mu = \mu_{custom}$
$= 800\ mPa \cdot s$
$= 800\ mPa \cdot s$
Dynamic viscosity (as entered)

[7] T_{line} = T (no T_{line} entered)
$= 20\ °C$
$= 20\ degC$
Design temperature of the line to insulate: the fluid temperature T, since no T_{line} was entered

[11] $Q_{design} = Q$
$= 1\ m^3/h$
$= 1\ m^3/h$
Design flow (as entered)

[12] v_{s1} = transfer line: $Q / (\pi D^2/4)$
$= 1\ m^3/h / (\pi \times 50^2\ mm^2 / 4)$
$= 0.14147\ m/s$
Velocity, transfer line

[13] Re_{s1} = transfer line: $\rho\ v\ D / \mu$
$= 888 \times 0.14147 \times 0.05 / 0.8$
$= 7.8516$
Reynolds number, transfer line

[14] f_{s1} = transfer line: Darcy friction factor
= 64 / Re (laminar)
$= 8.1512$
Friction factor, transfer line

[15] $h_{f,s1}$ = transfer line: $f (L + L_{eq})/D \cdot v^2/2g + K\ v^2/2g$
$= 8.1512 \times (40 + 0) / 0.05 \times 0.0010204 + 0 \times 0.0010204$
$= 6.6542\ m$
Head loss, transfer line

[16] $h_{f,total} = \Sigma\ h_f$
$= 1$ segment(s)
$= 6.6542\ m$
Total friction and fitting loss at the design flow

[18] $V_{pipe} = \Sigma\ \pi/4 \cdot D^2 \cdot L$
$= 1$ segment(s)
$= 78.54\ L$
Water held in the pipe segments

[19] $H_{static} = \Sigma\ dz + h_{out}$
$= 0 + 0$
$= 0\ m$
Static head

[20] $H_{sys} = H_{static} + h_{f,total}$
$= 0\ m + 6.6542\ m$
$= 6.6542\ m$
System head at the design flow

[21] $dp_{total} = \rho * 9.80665\ m/s^2 * h_{f,total}$
$= 888\ kg/m^3 \times 9.80665\ m/s^2 \times 6.6542\ m$
$= 57.947\ kPa$
Friction pressure drop at the design flow

[22] dp_{sys} = ρ * 9.80665 m/s^2 * H_{sys}
= 888 kg/m³ × 9.80665 m/s^2 × 6.6542 m
= 57.947 kPa
System pressure at the design flow
[23] P_{hyd} = ρ * 9.80665 m/s^2 * Q_{design} * H_{sys}
= 888 kg/m³ × 9.80665 m/s^2 × 1 m³/h × 6.6542 m
= 0.016096 kW
Hydraulic power at the design point
[26] v_{peak} = highest segment velocity
= 0.14147 m/s
= 0.14147 m/s
Highest velocity
[27] u_v = v_{peak} / v_{max}
= 0.14147 / 3
= 0.047157
Velocity utilisation

5. Results

SYMBOL	VALUE	DESCRIPTION
T_{line}	20 degC	Line design temperature
v_{s1}	0.14147 m/s	Velocity, transfer line
Re_{s1}	7.8516	Re, transfer line
f_{s1}	8.1512	f, transfer line
$h_{f,s1}$	6.6542 m	h_f, transfer line
$h_{f,total}$	6.6542 m	Σ h_f
H_{static}	0 m	Static head
V_{pipe}	78.54 L	Water held in the pipes
H_{sys}	6.6542 m	System head at Q
dp_{total}	57.947 kPa	Friction Δp
dp_{sys}	57.947 kPa	System Δp
Q_{design}	1 m³/h	Design flow
P_{hyd}	0.016096 kW	Hydraulic power
v_{peak}	0.14147 m/s	Highest velocity

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Velocity ≤ limit	u_v = 0.047157	u_v ≤ 1	0.0472	

7. Applicability

WARNING — input.circuit: circuit (Circuit (a closed flow-and-return loop has no static lift: the rises entered are ignored)) was not entered; the template default 'open (the pump lifts the water by the rises entered)' is used.
WARNING — input.n_ratio: n_{ratio} (Speed ratio n_2/n_1 (affinity)) was not entered; the template default 1 is used.
WARNING — input.D_ratio: D_{ratio} (Impeller diameter ratio D_2/D_1 (affinity; the law is chosen below)) was not entered; the template default 1 is used.

WARNING — input.D_law: D_{law} (Law for the diameter ratio: a trimmed impeller of the same casing (Q ∝ D, H ∝ D²) or a geometrically similar pump (Q ∝ D³, H ∝ D²)) was not entered; the template default 'impeller trim (Q ∝ n·D, H ∝ n²·D²)' is used.
WARNING — input.eta_pump: η_{pump} (Pump efficiency at duty (for shaft power)) was not entered; the template default 0.7 is used.
WARNING — input.p_atm: p_{atm} (Atmospheric pressure (NPSH)) was not entered; the template default 101.3 kPa is used.
WARNING — input.z_s: z_s (Static head at the pump suction (+ flooded, − lift)) was not entered; the template default 0 m is used.
WARNING — laminar: segments: transfer line is laminar (Re = 7.8516); f = 64/Re. The system curve and the duty point scale its friction with Q (h_f ∝ Q), not Q².

8. Verification

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Template bs.pipe-pump@1.8.1 Content hash 4723d62a61e80298 Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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9. References

1. Crane Co. Flow of Fluids Through Valves, Fittings and Pipe. Technical Paper No. 410. Crane Co.
2. Çengel YA, Cimbala JM. Fluid Mechanics: Fundamentals and Applications. 4th ed. McGraw-Hill Education; 2018.
3. Chartered Institution of Building Services Engineers. CIBSE Guide C: Reference Data. CIBSE; 2007.
4. Kell GS. Density, thermal expansivity, and compressibility of liquid water from 0° to 150°C: correlations and tables for atmospheric pressure and saturation reviewed and expressed on 1968 temperature scale. J Chem Eng Data. 1975;20(1):97-105.